

Software Engineering

Lecture 6b: Software Design & Object-Oriented Principles

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Focus Questions

- What is **Software Design** and how does it connect requirements to implementation?
- What are the **4 fundamental design activities** in software engineering?
- What are general design principles (**Abstraction, Encapsulation, Modularity**)?
- Why is **High Cohesion & Low Coupling** the golden rule of software architecture?
- What are the **SOLID Object-Oriented Design Principles** (SRP, OCP, LSP, ISP, DIP)?
- How can **AI & LLMs** assist developers in software design and code refactoring?

What is Software Design?

- **Definition:**

Software Design is the creative process of transforming customer requirements into a blueprint for building software components, modules, interfaces, and data structures.

- **The Bridge Role:**

- **Requirements Engineering** → *What the customer needs (System Specification).*
- **Software Design** → *How the software system is structured (Architectural Blueprint).*
- **Implementation & Coding** → *Constructing executable code.*

4 Fundamental Software Design Activities

1. Architectural Design:

- Identifying overall system structure, principal sub-systems, and communication pipelines.

2. Interface Design:

- Specifying precise contracts between sub-systems, REST APIs, and public class methods.

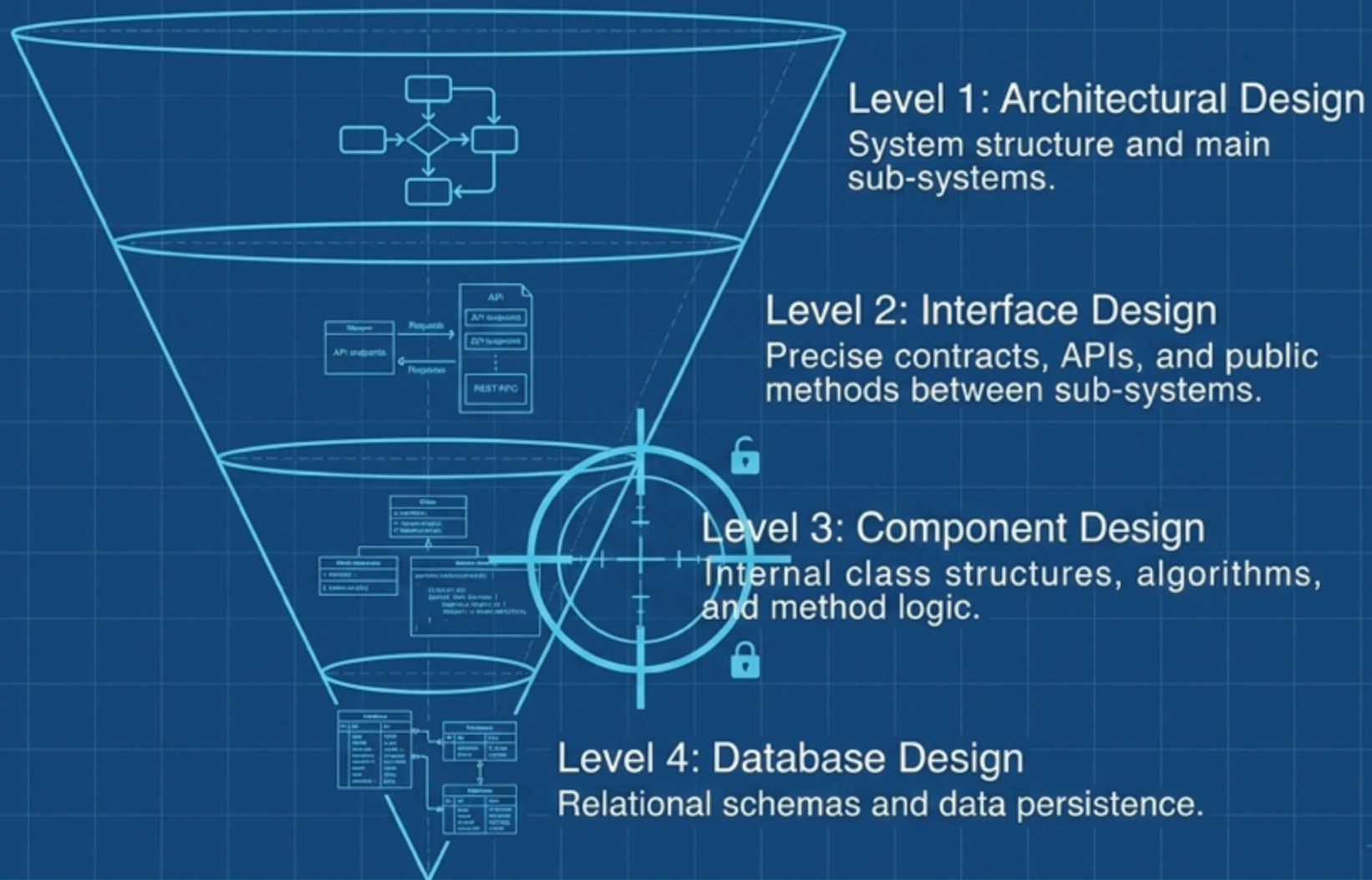
3. Component (Detailed) Design:

- Designing internal class structures, algorithms, data types, and method logic.

4. Database / Data Structure Design:

- Designing relational schemas, object graphs, or data persistence models.

Zooming In: The 4 Fundamental Design Activities



General Design Principles (Part 1)

- **1. Abstraction (抽象化):**
 - Hiding low-level implementation complexity behind clean high-level concepts.
 - Allows engineers to reason about complex systems without getting overwhelmed by detail.
- **2. Encapsulation & Information Hiding (封裝與資訊隱藏):**
 - Parnas's Principle: Bundling state and behavior inside a module, hiding internal variables behind public interfaces.
 - Prevents external code from introducing illegal state modifications.

General Design Principles (Part 2)

- **3. Modularity (模組化):**
 - Decomposing a large, monolithic software system into discrete, self-contained modules.
 - Enables parallel team development, independent testing, and partial reuse.
- **4. Separation of Concerns (SoC / 關注點分離):**
 - Splitting software into distinct features that overlap as little as possible (e.g. decoupling UI rendering from database access).

The Core Tenets of Software Design

Abstraction

Hiding low-level complexity behind clean, high-level concepts to prevent cognitive overload.

Encapsulation

Parnas's Principle—hiding internal state behind public interfaces to prevent illegal modifications.

Modularity

Decomposing monolithic code into discrete, testable, reusable units for parallel team development.

Separation of Concerns (SoC)

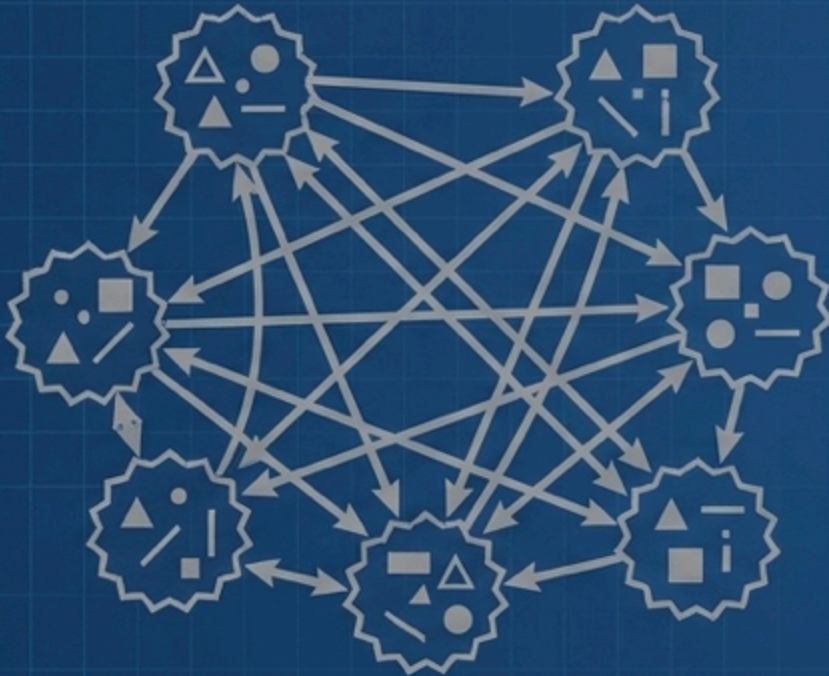
Splitting software into distinct, non-overlapping features (e.g., UI vs. database access).

High Cohesion vs. Low Coupling

[MICRO_LEVEL] // THE GOLDEN RULE

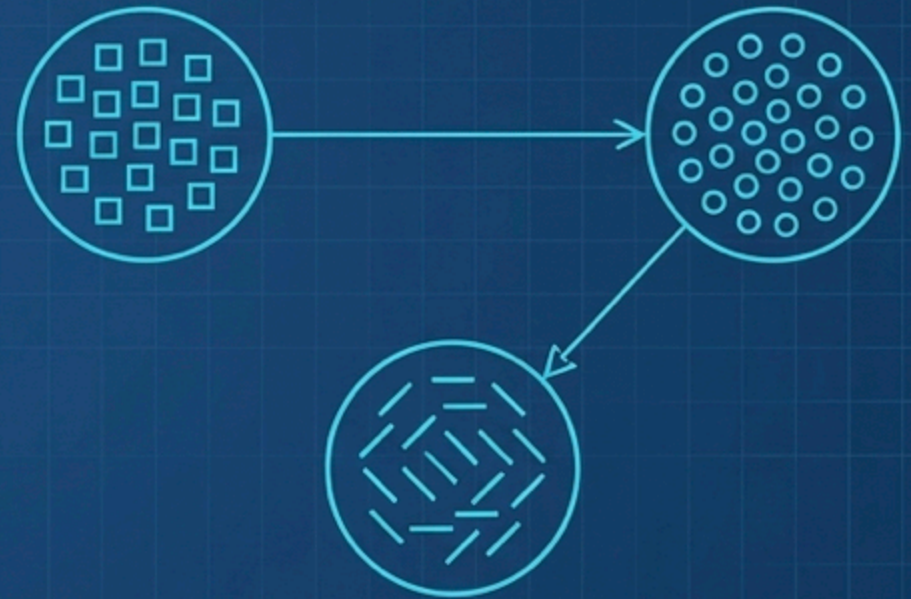
High Cohesion & Low Coupling

✗ Bad Design (Spaghetti Code)



DB changes break email logic!

✓ Good Design (The Golden Rule)



Components are independently testable and reusable.

Cohesion & Coupling Rules

- **Cohesion (凝聚度):**
 - Measures how strongly related the internal responsibilities inside a single module are.
 - **Goal: High Cohesion** (A class should focus on a single well-defined purpose).
- **Coupling (耦合度):**
 - Measures the degree of direct interdependence between separate modules.
 - **Goal: Low Coupling** (Modules communicate via abstract interfaces with minimal dependencies).
- **Golden Rule of Software Design:**

Maximize Cohesion, Minimize Coupling!

Cohesion & Coupling: Bad vs. Good Design

✗ Bad Design (Low Cohesion, High Coupling)

```
// Monolithic class doing EVERYTHING
class OrderManager {
    public void processOrder() {
        // 1. Direct MySQL DB logic
        // 2. Calculate discounts
        // 3. Render HTML invoice
        // 4. Send SMTP email directly
    }
}
```

- **Problem:** Hard to maintain; DB change breaks email logic!



✓ Good Design (High Cohesion, Low Coupling)

```
// Decoupled single-purpose classes
class OrderProcessor {
    private DiscountCalculator discount;
    private OrderRepository repo;
    private NotificationService notifier;
    // Communicates via interfaces
}
```

- **Benefit:** Each component is independently testable and reusable.

Concept Check Question 1

In software design, what does "Low Coupling" combined with "High Cohesion" achieve?

- **A.** It increases program execution speed by eliminating class methods.
- **B.** It creates highly modular code where changes to one module have minimal ripple effects on others.
- **C.** It forces all data fields to be global variables across sub-systems.
- **D.** It replaces the need for unit testing during development.



Concept Check Question 1: Answer

In software design, what does "Low Coupling" combined with "High Cohesion" achieve?

- **Correct Answer: B**
- **Explanation:** High Cohesion ensures a module has a single focused purpose, while Low Coupling minimizes inter-module dependencies, preventing cascading code changes.



Object-Oriented Design: The SOLID Principles

[MICRO_LEVEL] // OBJECT-ORIENTED FOUNDATIONS

The SOLID Principles: Visual Analogies

S

SRP



Every class handles a single responsibility.

O

OCP



Open for extension via ports, closed for internal modification.

L

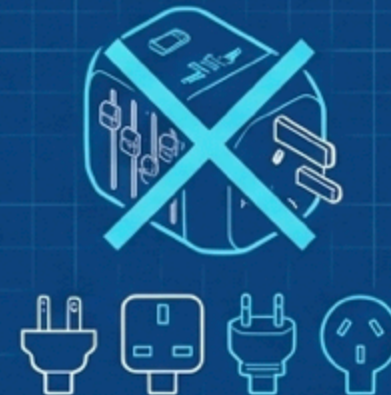
LSP



Subtypes must fulfill all base commitments.

I

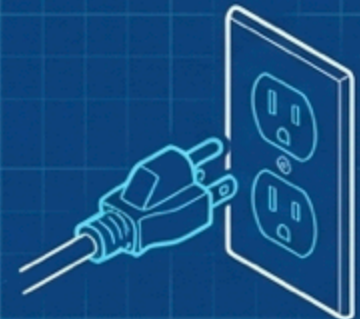
ISP



Create small, role-specific interfaces, not fat general ones.

D

DIP



Depend on abstractions (the socket), not concrete hardwiring.

S – Single Responsibility Principle (SRP)

- **Definition:**

A class should have one, and only one, reason to change.

- **Core Idea:**

- Every class should be responsible for a single part of the system's functionality.

- **Why SRP Matters:**

- When requirements change (e.g. database schema change or email format change), only the specific responsible class needs to be modified.

SRP: Bad Design vs. Good Design

✗ Bad Design (Violates SRP)

```
// "God Class" handling 3 distinct roles
class UserReportManager {
    public void fetchUserData() { /* DB */ }
    public void generatePdfReport() { /* PDF */ }
    public void sendEmailReport() { /* Email */ }
}
```

- Changes to PDF layout or Email server break the same class!



Good Design (Follows SRP)

```
// 3 Single-responsibility classes
class UserRepository {
    public User fetchUserData() { ... }
}

class PdfReportFormatter {
    public byte[] generatePdf(User u) { ... }
}

class EmailService {
    public void sendEmail(String to, byte[] pdf) { ... }
}
```

O – Open / Closed Principle (OCP)

- **Definition:**

Software entities should be open for extension, but closed for modification.

- **Core Idea:**

- You should be able to add new functionality without editing existing, tested source code.

- **How to Achieve:**

- Use **Polymorphism** and interfaces (e.g. Strategy Pattern) instead of `if-else` or `switch` blocks on types.

OCP: Bad Design vs. Good Design

✗ Bad Design (Violates OCP)

```
class DiscountCalculator {  
    public double calculate(String type, double price) {  
        if (type.equals("VIP")) return price * 0.8;  
        else if (type.equals("STUDENT")) return price * 0.9;  
        // Adding new tier REQUIRES editing this class!  
        return price;  
    }  
}
```



Good Design (Follows OCP)

```
interface DiscountStrategy {  
    double apply(double price);  
}  
class VipDiscount implements DiscountStrategy {  
    public double apply(double p) { return p * 0.8; }  
}  
class StudentDiscount implements DiscountStrategy {  
    public double apply(double p) { return p * 0.9; }  
}  
// Add new tier by adding NEW class!
```

L – Liskov Substitution Principle (LSP)

- **Definition:**

Subtypes must be substitutable for their base types without altering the correctness of the program.

- **Core Idea:**

- Derived classes must fulfill all commitments and invariants of the base class.

- **Why LSP Matters:**

- If a subclass throws unexpected exceptions or breaks base assumptions, polymorphic code fails at runtime.

LSP: Bad Design vs. Good Design

✗ Bad Design (Violates LSP)

```
class Rectangle {
    protected int w, h;
    public void setWidth(int w) { this.w = w; }
    public void setHeight(int h) { this.h = h; }
}
class Square extends Rectangle {
    // Overrides setWidth to change height too!
    public void setWidth(int w) { this.w = w; this.h = w; }
}
// Client code expecting Rectangle behavior FAILS!
```



(Follows LSP)

Good Design

```
interface Shape {
    int getArea();
}
class Rectangle implements Shape {
    private int w, h;
    public int getArea() { return w * h; }
}
class Square implements Shape {
    private int side;
    public int getArea() { return side * side; }
}
```

I – Interface Segregation Principle (ISP)

- **Definition:**

Clients should not be forced to depend on methods they do not use.

- **Core Idea:**

- Avoid creating "fat" general-purpose interfaces. Instead, create many small, specific role interfaces.

- **Why ISP Matters:**

- Prevents dummy/unsupported method implementations throwing `UnsupportedOperationException`.

ISP: Bad Design vs. Good Design

✗ Bad Design (Violates ISP)

```
interface MultifunctionDevice {
    void print();
    void scan();
    void fax();
}
// BasicPrinter forced to implement scan & fax!
class SimplePrinter implements MultifunctionDevice {
    public void print() { /* OK */ }
    public void scan() { throw new Error(); }
    public void fax() { throw new Error(); }
}
```



Good Design (Follows ISP)

```
interface Printer { void print(); }
interface Scanner { void scan(); }
interface Fax { void fax(); }

class SimplePrinter implements Printer {
    public void print() { /* Clean */ }
}

class AllInOnePrinter implements
    Printer, Scanner, Fax { ... }
```

D – Dependency Inversion Principle (DIP)

- **Definition:**

High-level modules should not depend on low-level modules. Both should depend on abstractions.

- **Core Idea:**

- Depend on **Interfaces**, not concrete classes. Use **Dependency Injection (DI)**.

- **Why DIP Matters:**

- Allows swapping database engines, HTTP clients, or 3rd-party services without rewriting business logic.

DIP: Bad Design vs. Good Design

✗ Bad Design (Violates DIP)

```
class OrderService {  
    // Tightly coupled to concrete MySQL class  
    private MySQLDatabase db = new MySQLDatabase();  
  
    public void saveOrder(Order o) {  
        db.insert(o); // Cannot mock for testing!  
    }  
}
```



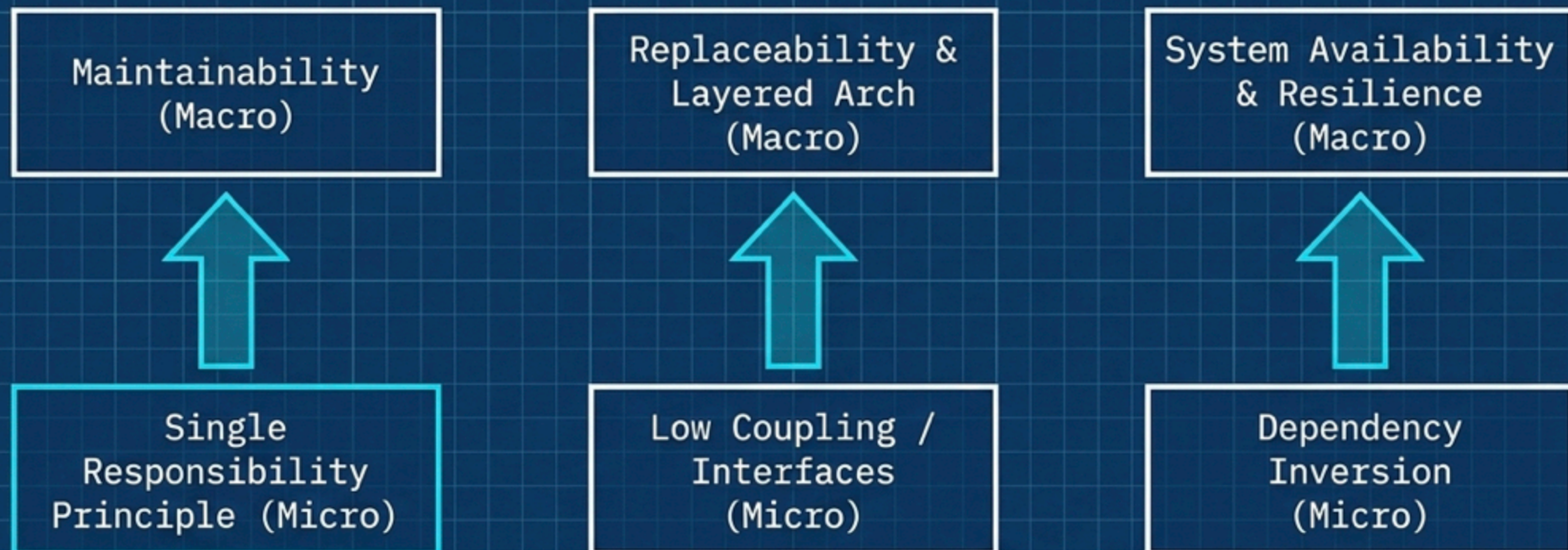
Good Design (Follows DIP)

```
interface Database { void insert(Order o); }  
  
class OrderService {  
    private Database db; // Depends on abstraction  
    // Injected via Constructor (DI)  
    public OrderService(Database db) {  
        this.db = db;  
    }  
}
```

SOLID: The Good vs. Bad Matrix

Principle	✗ The Violation	✓ The Fix	Core Benefit
SRP	Changes to PDF layout and Email server break the same class.	Separate PDF and Email into distinct classes.	Isolates requirement changes.
OCP	Using massive if-else or switch blocks on object types.	Use Polymorphism and interfaces (Strategy Pattern).	Add features without editing tested code.
LSP	Subclasses throwing unexpected exceptions breaking base assumptions.	Derived classes strictly fulfill base invariants.	Prevents polymorphic runtime crashes.
ISP	Dummy method implementations throwing <code>UnsupportedOperationException</code> .	Segregate into multiple small, focused interfaces.	Prevents forced dependency on unused methods.
DIP	High-level business logic instantiates concrete database engines.	Use Dependency Injection (DI) and depend on interfaces.	Allows swapping services without rewriting logic.

The Micro-Macro Harmony Matrix



Insight: Macro-architectural blueprints are useless without Micro-level adherence. Code design is where architecture becomes reality.

[AI_COPILOT] // INTELLIGENT AUGMENTATION

The AI Architect Copilot Dashboard

Macro / Architecture Automation

- **Pattern Recommendation:** Analyzing non-functional requirements to suggest Microservices vs. Layered.
- **ATAM Analysis:** Evaluating conflicting attributes (Security vs. Latency).
- **Threat Modeling:** Scanning diagrams for Single Points of Failure (SPOF) and STRIDE threats.

Micro / Software Design Automation

- **SOLID Refactoring:** Identifying God Classes and suggesting factory patterns.
- **Code Smell Detection:** Highlighting tight coupling and duplicate logic.
- **Interface Generation:** Auto-generating OpenAPI specs, gRPC .proto files, and decoupled mocks for TDD.

4 Core AI Applications in Software Design

1. Automated SOLID Refactoring:

- Identifies God Classes (SRP violations) and suggests Strategy or Factory patterns to achieve OCP.

2. Code Smell Detection:

- Scans legacy code for tight coupling, duplicate logic, shotgun surgery, and feature envy.

3. Design Pattern Recommendation:

- Recommends GoF design patterns (Observer, Decorator, Adapter, Builder) for recurring design challenges.

4. Interface Contract & Mock Generation:

- Auto-generates decoupled interfaces and mock objects for Test-Driven Development (TDD).

Human-in-the-Loop: Design Risks & Best Practices

- **Risks of Unchecked AI in Software Design:**
 - **Over-Engineering:** AI suggesting complex Gang of Four patterns for simple functions.
 - **Superficial Refactoring:** Renaming variables without resolving fundamental architectural flaws.
- **The Golden Rule:**

AI Proposes Refactoring; Developers Validate Software Quality!

Engineers must ensure that refactored designs actually simplify code readability and long-term maintainability.

Human-in-the-Loop: Risks & Blind Spots



Risk 1: Architectural Blind Spots

AI lacks context on team tech-stack familiarity, cloud vendor costs, and organizational culture.



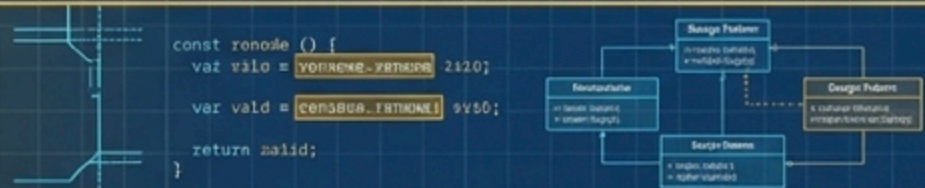
Risk 2: Premature Microservices

The danger of AI recommending complex distributed architectures for simple CRUD applications.



Risk 3: Superficial Refactoring

AI renaming variables or over-engineering with GoF patterns without fixing fundamental architectural flaws.



Golden Rule Callout: AI suggests. The Architect decides.
AI evaluates code, humans evaluate context.

Concept Check Question 2

Which SOLID principle states that you should be able to add new system features by adding new classes without modifying existing tested code?

- **A.** Single Responsibility Principle (SRP)
- **B.** Open/Closed Principle (OCP)
- **C.** Interface Segregation Principle (ISP)
- **D.** Liskov Substitution Principle (LSP)



Concept Check Question 2: Answer

Which SOLID principle states that you should be able to add new system features by adding new classes without modifying existing tested code?

- **Correct Answer: B**
- **Explanation:** The Open/Closed Principle (OCP) requires systems to be open for extension (via polymorphism/interfaces) but closed for modification.



Recap: Fill-in-the-blank Quiz

Test your understanding of the core concepts in this chapter:

1. ____ measures how focused internal module responsibilities are, while ____ measures inter-module dependencies.
2. The ____ Responsibility Principle states a class should have only one reason to change.
3. The ____ Substitution Principle requires subtypes to be substitutable for base types without errors.
4. Dependency ____ means high-level business logic should depend on abstract interfaces, not concrete classes.



Recap: Answers & Summary

Here are the completed concepts:

1. **Cohesion** measures how focused internal module responsibilities are, while **coupling** measures inter-module dependencies.
2. The **Single** Responsibility Principle states a class should have only one reason to change.
3. The **Liskov** Substitution Principle requires subtypes to be substitutable for base types without errors.
4. Dependency **Inversion** means high-level business logic should depend on abstract interfaces, not concrete classes.



References

- **Sommerville Software Engineering Book (Chapter 7 - Design and Implementation)**
 - [Martin Fowler Refactoring & Clean Code Guide](#)
 - [Robert C. Martin \(Uncle Bob\) SOLID Principles](#)